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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
FOURTH SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCZIM201 - INDUSTRIAL MICROBIOLOGY AND FERMENTATION TECHNOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)
and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Fill in the blanks . Answer all questions
Each question carries 1 marks

1. Precipitation helps in concentrating and..... the product.[K]/[C01]
Ans:Purifying
2. A typical fermenter is usually constructed using .....steel to avoid corrosion.
Ans:Stainless **[K]/[CO2]**
3. In brewing, proteases help in..... clarification. [K]/[CO4]
Ans:Beer
4. Using microbes for waste treatment is preferred because they reduce environmental.....K]/[CO1]
Ans:Pollution
5. Acetic acid bacteria grow best at.....pH[K]/[CO2]
Ans:Acidic
6. Cellulase activity is affected by..... and.....[K]/[CO4]
Ans:Temperature and pH
7. Selective media help isolate microbes with..... characteristics.[K]/[CO1]
Ans:Specific
8. SSF generally produces more concentrated products due to low ..... volume.[K]/[CO2]
Ans:moisture
9. Ferric iron acts as an .....agent in uranium bioleaching.[U]/[CO5]
Ans:oxidizing
10.The equipment used for submerged fermentation is called a .....[K]/[CO1]
Ans:fermenter (or bioreactor). [1x10 = 10]

Part B
Short Answer Questions. Answer any 10 questions
Each question carries 2 marks

11. Define continuous sterilization.[K]/[C01]
Continuous sterilization is a process where medium is sterilized while flowing through a system.
It uses heat exchangers for rapid heating and cooling.This method is energy efficient and suitable for large-scale production.It reduces contamination risk compared to batch sterilization.

12. Discuss the nutrient concentration in batch fermentations.[K]/[CO2
Nutrients are added at the beginning in fixed amounts.Their concentration decreases as microbes grow.Limiting nutrients can control growth and product formation.Proper balance is essential for maximum yield.

13. State how are aerobic conditions essential for glutamic acid production.[K]/[CO4]
Glutamic acid production requires high oxygen supply.Oxygen helps in efficient metabolism of carbon sources.Insufficient oxygen reduces product yield.Hence, proper aeration and agitation are essential.

14. Explain the exponential phase preferred for inoculum transfer.[K]/[C01]
Cells in exponential phase are highly active and viable.They adapt quickly to new medium conditions.
This reduces lag phase in fermentation.Thus, it ensures rapid growth and productivity.

15. Explain how does over fermentation lead to poor texture.[U]/[CO3]
Over fermentation leads to excess acid or enzyme activity.This breaks down structure of the product.
It causes soft, undesirable texture.Hence, proper timing is important.

16. Discuss how is MEOR applied in mature oil fields.[A]/[CO5]
Microbes are injected into oil reservoirs.They produce gases, acids, or surfactants.These help in mobilizing trapped oil.Thus, oil recovery efficiency increases.

17. Compare chemical and physical methods of immobilisation.[K]/[C01]
Chemical methods use covalent bonding or cross-linking.Physical methods include adsorption and entrapment.Chemical methods are stronger but may affect cells.Physical methods are simpler and less harmful.

18. Explain the creation circulation between riser and downcomer zones.[K]/[CO4]
Air is introduced in the riser section.This reduces density and causes upward movement.
Liquid flows down through the downcomer.This creates continuous circulation in the system.

19. Explain how is pH controlled during penicillin fermentation.[K]/[CO2]
pH is maintained using buffers or alkali addition.Acid formation lowers pH during fermentation.

Automatic pH control systems are used.Optimal pH ensures maximum penicillin yield.

20. Define rate of dilution in continous fermenter.[K]/[CO2]
Dilution rate is the rate of fresh medium addition.It is equal to flow rate divided by culture volume.
It controls microbial growth rate.High dilution can wash out cells.

21. Explain the causes of formation in fermentation.[K]/[C01]
Foam is caused by proteins and agitation.Air bubbles get stabilized by surface-active agents.Excess foam affects oxygen transfer.Antifoaming agents are used to control it.

22. List the measures ensure consistant quality in oriental fermented products.[U]/[CO3]
Use of pure starter cultures ensures consistency.Maintain hygiene and controlled conditions.
Standardize raw materials and process steps.Regular quality checks improve product stability.

[2x10 = 20]

Part C
Short Essay Questions.Answer any 5 questions
Each question carries 4 marks

23. **Discuss the limitations of copper bioleaching. [U]/[CO5] (Any four limitation-1 mark each)**

- Slow rate of copper extraction
- Low recovery efficiency
- Risk of acid mine drainage and pollution
- Requires strict acidic pH and temperature conditions
- Limited to sulfide ores only

24. **Explain how traditional fermentation practices evolved into modern industrial fermentation techniques. [K]/[CO1]**
(Traditional and modern ferementaion-3 mark,Discoveries by Louis Pasteur-1 mark)
Traditional fermentation began as a natural process using wild microorganisms for making foods like curd, wine, and bread. Over time, people learned to control conditions such as temperature and cleanliness to improve consistency. Scientific discoveries by Louis Pasteur proved that microorganisms are responsible for fermentation. This led to the isolation and use of specific microbial strains for better yield and quality. Modern industrial fermentation uses sterile conditions, controlled pH, temperature, and oxygen in large bioreactors. It is now widely applied for producing antibiotics, enzymes, alcohol, and other commercial products efficiently.
25. Describe how to prevent contamination during cheese aging.[K]/[CO2]
(Pasteurized milk and hygiene-2 mark,Temperature-1 mark,sterilization-1 mark)
Use high-quality pasteurized milk and maintain strict hygiene during handling.
Sterilize all equipment, aging rooms, and shelves before use.

Control temperature and humidity to optimal levels for proper ripening.
Limit exposure by handling cheese with clean tools and minimal contact.
Encourage beneficial cultures to out-compete harmful microorganisms.
Regularly inspect cheeses and remove any contaminated or mold-infected samples promptly.

26. **Explain how does agitation affect morphology of streptomyces and antibiotic yield.[U]/[CO4]**
(Effect of agitation on Morphology-2 Marks,Effect on Antibiotic yield- 2 Marks)

- Affects oxygen transfer and nutrient mixing
- Low agitation forms large mycelial clumps
- High agitation produces smaller pellets/fragments
- Excess agitation damages hyphae
- Optimal agitation gives proper pellet size

27. **Differentiate between positive and negative selection techniques.[K]/[CO1]**
Any 4 differences with one example-1mark each

- Positive selection allows growth of desired cells under selective conditions.
- Negative selection eliminates unwanted cells while desired cells survive
- Positive selection uses markers like antibiotic resistance
- Negative selection uses toxic agents to kill specific cells
- Positive selection directly identifies transformed/recombinant cells
- Negative selection removes non-recombinant or undesired cells

28. Compare ale and larger fermentation.[K]/[CO2](Definition-1mark,Type-1 mark,Any 2 differences-1mark each)

- Ale fermenter is small-scale; large fermenter is industrial-scale
- Ale fermenter used for beer production; large fermenter for antibiotics, enzymes, alcohol
- Ale fermenter mainly batch; large fermenter can be batch/fed-batch/continuous
- Ale fermenter has simpler design; large fermenter is complex

29. Summarize the role of microbial selecion in gold bioleaching.[K]/[CO5]
(Bioleaching definition-1mark,Organism-1mark,Role-2 mark)

Microbial selection ensures the use of efficient metal-solubilizing microbes like Acidithiobacillus ferrooxidans.Selected strains show high tolerance to toxic metals and acidic conditions.
They enhance oxidation of sulfide minerals associated with gold.This improves gold exposure and recovery efficiency.Proper selection increases process stability and consistency.

[4x5 = 20]